CITY COLLEGE

Approved by AICTE, New Delhi. Recognized by Govt. of Karnataka. Affiliated to Bangalore University, accredited by NAAC with "B+" Grade No.27/2, 33rd Cross, 2nd Main Rd, 7th Block,Jayanagar, Bengaluru, Karnataka 560070 Phone: 6364914155 / 56,

+918026649277 Website: www.citycollege.ac.in Email: admin@citycollege.ac.in

Project Report On

"Japan Aerospace Exploration Agency (JAXA) Cyber Incident (2024)"

Submitted in partial fulfilment of the requirement for the project work for the degree of

(BACHELOUR OF COMPUTER APPLICATION) BCA

2025–26

BANGALORE UNIVERSITY

Under the Guidance Of

Ms. Chandana Khatavakar

Assistant Professor

Department of Computer Application

Submitted by

UMESH R

(U03BX24S0108)

CITY COLLEGE, JAYANAGAR, BENGALURU

DEPARTMENT OF COMPUTER APPLICATION

CERTIFICATE

This is to certify that the project synopsis entitled “Japan Aerospace Exploration Agency (JAXA) Cyber Incident (2024)” is submitted by Mr Umesh R(USN: U03BX24S0108) in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) from Bangalore University for the academic year 2025–2026.

The project synopsis has been prepared under the guidance and supervision of the undersigned. The synopsis is submitted for academic evaluation in accordance with the applicable department and university guidelines.

_____	_____	_____
Project Guide	**Head of the Department**	**Principal**
Ms. Chandana Khatavakar	Dr. Nageshwari	**City College**
Assistant Professor	Department of Computer Application	Jayanagar, Bengaluru

Place: Bengaluru

Date: 11-10-2026

Abstract

The Japan Aerospace Exploration Agency (JAXA) experienced a significant cybersecurity incident involving unauthorized access to part of its internal network. JAXA detected the incident in October 2023 following a notification from an external organization and publicly disclosed details of the incident in July 2024. According to JAXA, attackers appear to have exploited a vulnerability in a VPN device to gain initial access to some internal servers and computers. The attackers subsequently expanded their access, stole account-related information and used the stolen credentials to access Microsoft 365 while impersonating legitimate users. JAXA confirmed that some information under its management was leaked, including information related to joint work with external organizations and personal information. However, JAXA stated that the compromised systems did not contain sensitive information concerning rocket or satellite operations. JAXA responded by isolating affected systems, blocking malicious communications, conducting forensic investigations and working with security organizations. The incident demonstrates the importance of vulnerability management, network segmentation, identity protection, monitoring and rapid incident response for organizations involved in aerospace and other sensitive activities.

Introduction

Japan Aerospace Exploration Agency (JAXA) is Japan's national aerospace research and development organization. Its activities include space exploration, satellites, launch vehicles and aerospace research.

Because aerospace organizations handle valuable technical, personal and collaborative information, they are attractive targets for cyberattacks.

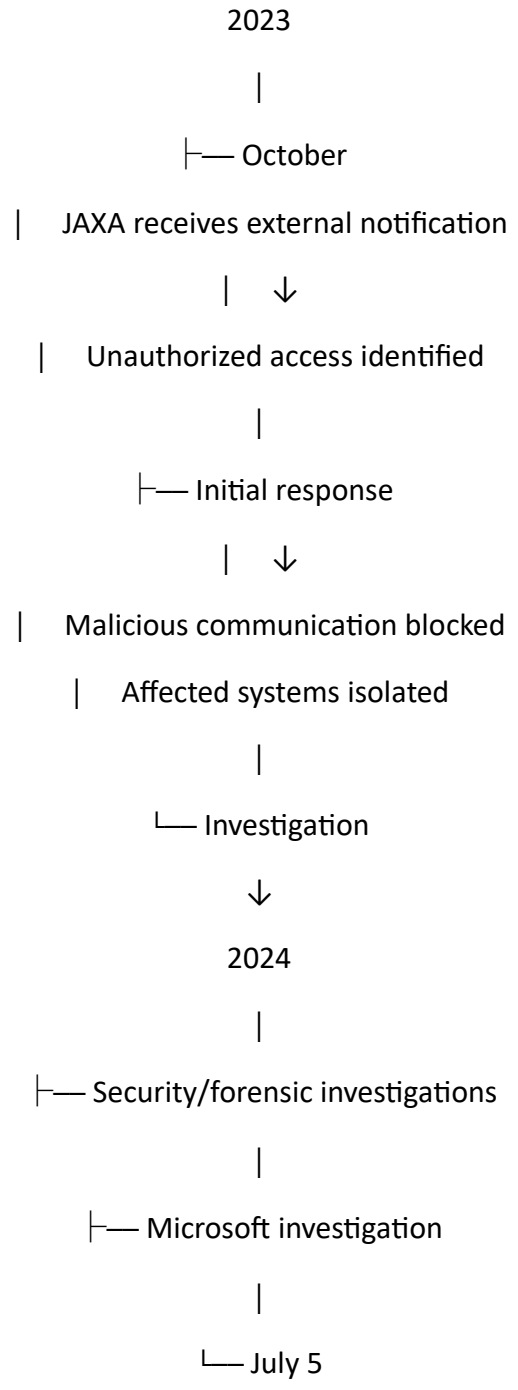
The JAXA incident is an important cybersecurity case because the attackers did not simply target a single computer. The intrusion involved:

VPN → Internal systems → Account information → Microsoft 365 → Information leakage

JAXA reported that the compromised systems did **not** handle sensitive information related to rocket and satellite operations. Nevertheless, the incident created concerns regarding information security and trust with external partners.

Incident Description

Timeline



JAXA publicly announces information leakage

JAXA stated that the incident was detected in **October 2023** based on information from an external organization. Security specialists and vendors were then involved in investigating the incident.

Important: The 2024 date refers mainly to JAXA's public disclosure and subsequent investigation/response reporting.

Attack Type / Vulnerability

The incident involved several stages of unauthorized access.

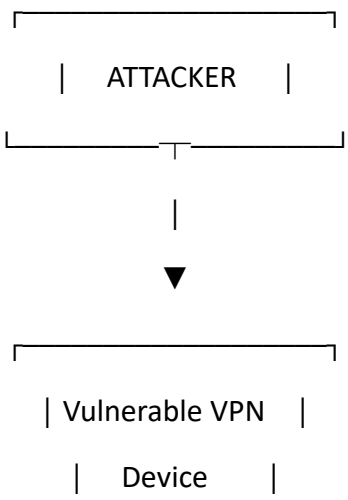
Primary attack characteristics:

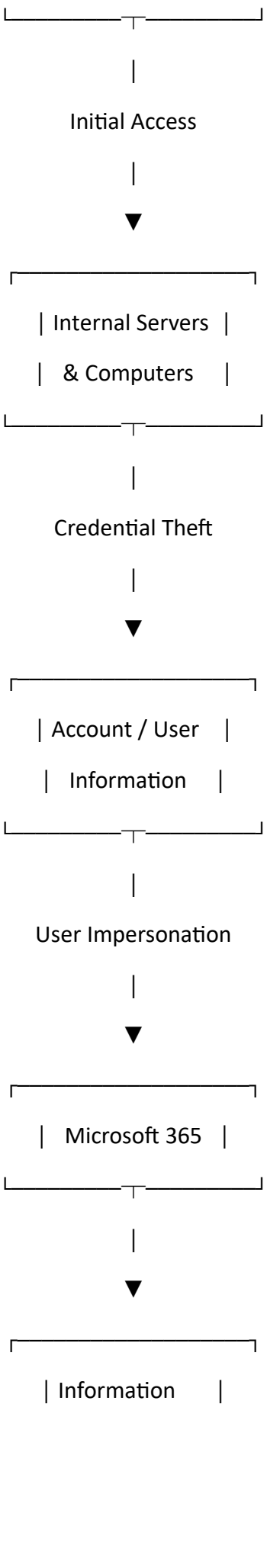
1. **VPN vulnerability exploitation**
    - Attackers used a vulnerability in a VPN device as the initial entry point.
2. **Network intrusion**
    - Attackers accessed some internal servers and computers.
3. **Credential theft**
    - Account-related information was stolen.
4. **Credential-based access**
    - Stolen information was used to access Microsoft 365.
5. **User impersonation**
    - Attackers accessed Microsoft 365 while appearing to be legitimate users.
6. **Data leakage**
    - Some information managed by JAXA was exposed.

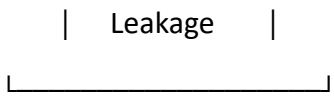
JAXA stated that a previously disclosed vulnerability was **highly likely** to have been exploited.

Attack Flow Diagram / Infographic

Use this as your required diagram in the PPT:




Initial Access
Internal Servers & Computers
Credential Theft
Account / User Information
User Impersonation
Microsoft 365
Information



This flow is based on JAXA's description of the intrusion sequence.

Root Cause Analysis

The main technical cause was the exploitation of a vulnerability in a **VPN device**.

However, the incident involved more than one weakness.

Root causes and contributing factors:

- Vulnerable external-facing VPN infrastructure
- Potential delay between vulnerability disclosure and remediation
- Compromise of internal systems after initial entry
- Theft of account information
- Ability to use stolen credentials against Microsoft 365
- Need for stronger monitoring of internal communications
- Difficulty detecting some malicious activity because previously unknown malware was identified during the investigation

JAXA reported that multiple unknown malware samples were discovered during the investigation, which made detection more difficult.

Impact Analysis

The incident resulted primarily in **information-security and trust-related impacts**.

Information impact:

- Some JAXA-managed information was leaked.
- Information associated with joint activities with external organizations was affected.
- Personal information was also involved.

Operational impact:

- JAXA reported that it did not identify significant disruption to related organizations' business activities.

- The affected systems did not contain sensitive rocket or satellite operational information.

Organizational impact:

- Potential damage to trust between JAXA and external partners.
- Additional investigation and cybersecurity costs.
- Requirement for stronger security controls.

JAXA explicitly described the incident as serious because it could undermine trust, even though critical rocket and satellite operational systems were not affected.

Detection and Response

JAXA's response included multiple stages.

1. Initial detection

- External notification helped JAXA identify unauthorized access.

2. Communication blocking

- JAXA blocked communication with malicious destinations.

3. System isolation

- Affected servers and devices were disconnected from the network.

4. Forensic investigation

- Security vendors investigated compromised servers and endpoints.
- Malware was identified and removed.

5. Microsoft 365 investigation

- Because unauthorized access to Microsoft 365 was suspected, Microsoft's specialist team conducted additional investigation.

6. External coordination

- JAXA worked with police, JPCERT/CC and IPA during the response.

These actions helped determine the scope of the compromise and reduce the possibility of continued unauthorized access.

Security Measures & Mitigation

JAXA introduced both short-term and long-term security measures.

Short-term measures:

- Faster vulnerability response
- Blocking suspicious communications
- Network isolation
- Malware investigation and removal
- Strengthened internal communications monitoring

Long-term measures:

- Improved vulnerability-management processes
- Stronger monitoring across networks and endpoints
- Improved external connection methods
- Stronger operational management
- Improved anti-spoofing measures
- Strengthened incident-response capabilities

JAXA's later reporting also describes strengthened monitoring, improved external connection methods and enhanced anti-spoofing measures as part of its efforts to prevent recurrence.

Lessons Learned

The JAXA incident provides several important cybersecurity lessons.

Lesson 1: Patch external-facing systems quickly
VPN devices can become a major entry point into an organization.

Lesson 2: Protect credentials
Stealing account information can allow attackers to move from internal systems to cloud services.

Lesson 3: Do not rely only on perimeter security
Even after attackers enter through a VPN, internal monitoring and segmentation can limit their movement.

Lesson 4: Cloud accounts require strong protection
Microsoft 365 and other cloud platforms should be protected with strong identity controls.

Lesson 5: Early detection matters
External notification played an important role in identifying the incident.

Lesson 6: Cooperation improves incident response
JAXA's coordination with specialist organizations helped investigate the attack.

Recommendations

Based on the incident, organizations can adopt the following recommendations:

1. **Continuous vulnerability scanning**
2. **Rapid patching of VPN and internet-facing systems**
3. **Multi-factor authentication for important accounts**
4. **Strong password and credential-management policies**
5. **Network segmentation**
6. **Endpoint Detection and Response (EDR)**
7. **Centralized security logging and monitoring**
8. **Continuous monitoring of Microsoft 365/cloud accounts**
9. **Least-privilege access**
10. **Regular cybersecurity awareness training**
11. **Incident-response exercises**
12. **Threat-intelligence sharing with trusted organizations**

Japan's national cybersecurity strategy also emphasizes vulnerability identification, improved government response capabilities and stronger collaboration.

Why This Case Was Selected

Reason for choosing the JAXA case:

The JAXA incident was selected because it demonstrates how a vulnerability in an external-facing network device can become the starting point for a broader cyber intrusion. It is particularly relevant to modern cybersecurity because the attack progressed from a VPN device to internal systems, credential theft and cloud-based services. The case also shows

that cybersecurity is not only about protecting confidential technical information; personal information, partner information and organizational trust can also be affected. Furthermore, the incident provides a real-world example of vulnerability management, forensic investigation, incident response and cooperation with cybersecurity organizations.

Conclusion

The JAXA cyber incident demonstrates the growing importance of cybersecurity for aerospace and research organizations.

The attackers reportedly used a vulnerability in a VPN device to gain initial access. They then expanded their access, obtained account-related information and used those credentials to access Microsoft 365. Some information managed by JAXA was leaked, including personal information and information connected with external collaborative activities.

Although JAXA stated that the compromised systems did not contain sensitive rocket or satellite operational information, the incident still represented a significant security and trust issue.

The most important conclusion is that cybersecurity must be treated as a **continuous process**. Organizations should combine vulnerability management, strong authentication, network segmentation, endpoint monitoring, cloud security, threat intelligence and effective incident-response procedures.

The JAXA case shows that even organizations involved in highly advanced technology can be affected by weaknesses in ordinary IT infrastructure. Therefore, protecting the basic security layer—such as VPNs, credentials and cloud accounts—is essential to protecting larger missions and organizations.

References

Use **at least these 6 credible references** in your final PPT/PDF:

1. **JAXA — Unauthorized Access and Information Leakage, July 5, 2024**
   JAXA official incident announcement
2. **JAXA — President's July 2024 Press Conference**
   JAXA President's July 2024 Press Conference
3. **JAXA — 2024 Sustainability Report**
   JAXA Sustainability Report
4. **JAXA — FY2024 Financial Report / Security Measures**
   JAXA Financial Report

5. **Japan National Center of Incident readiness and Strategy for Cybersecurity (NISC) — Cybersecurity 2024**
   Cybersecurity 2024 Overview
6. **National Cybersecurity Office of Japan — Cybersecurity policy and incident reporting**
   National Cybersecurity Office of Japan

Required Disclaimer

Disclaimer: This case study is prepared for academic and educational purposes. The information presented is based on publicly available reports and official disclosures concerning the JAXA cybersecurity incident. Some technical details of the incident have not been publicly disclosed by JAXA and therefore are not assumed or presented as confirmed facts. The analysis and recommendations in this report are the author's academic interpretation and should not be considered an official statement by JAXA or any government organization.

Suggested publication note

For your **low-plagiarism requirement**, don't copy paragraphs directly from JAXA or news articles. The report above is structured and worded as an **original academic analysis**, while the factual incident details are attributed to the official sources. Keep the references and citations in your final PDF.

Approximate report length: 1,600 words excluding references and diagram, which fits your **1,200–2,000-word** requirement.